

AI History & State of the Art

Outline

I. A brief history of AI

II. The state of the art

I. Brief History of AI

- 1943 McCulloch & Pitts: model of artificial neurons
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 McCarthy, Minsky, Newell, Simon, Shannon, Nash et al.
Dartmouth workshop: birth of "Artificial Intelligence"
- 1952-69 Early enthusiasm, great expectations, optimism fueled by early success on some problems thought to be hard
- 1966-73 Collapse in AI research: Progress was slower than expected.
Unrealistic predictions, Herbert Simon (1957)
AI discovers computational complexity.

Brief History of AI (cont'd)

- 1969-86 Expert systems
- 1980- AI becomes an industry: expert systems booms, then busts (1988-93): “AI Winter”
- 1986- Neural networks regain popularity
- 1987- Probabilistic reasoning and machine learning

Brief History of AI (cont'd)

- 1995- Emergence of intelligent agents
 - ♣ AI technologies continue to find applications in
 - ◆ information retrieval
 - ◆ data mining and knowledge discovery
 - ◆ customizable software systems
 - ◆ smart devices (e.g., home, automobiles)
 - ◆ agile manufacturing systems
 - ◆ autonomous vehicles
 - ◆ bioinformatics
 - ◆ internet tools: search engines, recommender systems
 - ◆ ...
 - ♣ Steady progress on fundamental AI research problems continues.

Brief History of AI (cont'd)

- 2001- Big data
 - e.g., ImageNet
- 2011- Deep learning
 - ♣ Successful large-scale real-world applications in
 - ♦ image recognition
 - ♦ natural language processing
 - ♦ speech recognition
 - ♦ machine translation
 - ♦ ...
 - ♣ Convolutional neural networks (CNNs)
 - ♣ Chat Generative Pre-trained Transformer (ChatGPT)

Turing Award Winners in AI

1969 Marvin Minsky (MIT)

1971 John McCarthy (Stanford)

National Medal of Science (1990)

1975 Allen Newell* (Carnegie Mellon)

National Medal of Science (1992)

Herbert Simon* (Carnegie Mellon)

Nobel Prize in Economics (1978)

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1994 Edward Feigenbaum (Stanford)

Raj Reddy (Carnegie Mellon)

2011 Judea Pearl (UCLA)

2018 Yoshua Bengio (U Montreal)

Geoffrey Hinton (U Toronto)

Nobel Prize in Physics (2024)

Yann Lecun (NYU & Facebook)

2024 Andrew Barto** (U Massachusetts)

Richard Sutton** (U Alberta)

*omitted from the textbook on p. 17.

Big mistake!

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[US News rankings of CS programs in AI \(2024\):](#)

1. Carnegie Mellon
2. MIT
3. Stanford
4. Berkeley

...

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Academic Genealogy

Solomon Lefschetz

Albert Tucker



Herbert Simon



John McCarthy



Marvin Minsky



Allen Newell



Edward Feigenbaum



Raj Reddy

II. The State of the Art

- Game playing
 - ♦ Chinook defeated human checkers champions (1994).
 - ♦ Deep Blue (IBM) defeated the reigning world chess champion Garry Kasparov (1997).
 - ♦ Supercomputer Watson (IBM) beat human champions on “Jeopardy” (2011).
 - ♦ AlphaGo (Google) beat the world’s No.1 ranked player Ke Jie in Go (2017).
 - ♦ Libratus (Carnegie Mellon) defeated the world’s best Texas Hold ‘em poker players head-to-head (2017), six-player (2019).
 - ♦ “AlphaStar” AI (Google) defeated human pros at StarCraft II (2019)

(cont'd)

- Robotic vehicles

- ◆ Stanley (Stanford) won the DARPA Grand Challenge (2005)
- ◆ Boss (Carnegie Mellon) won the DARPA's Urban Challenge (2007)
- ◆ Self-driving cars by Tesla, Google, etc.



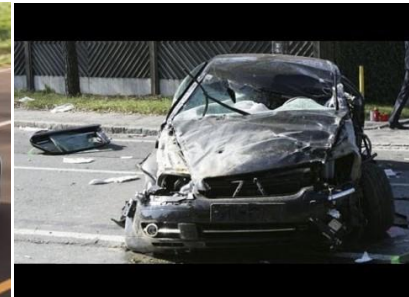
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Google car



Tesla car crash
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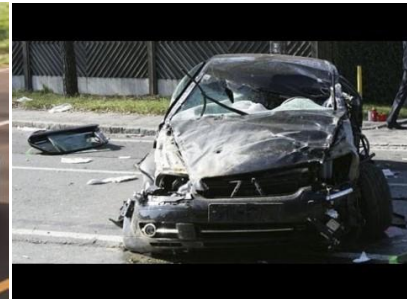
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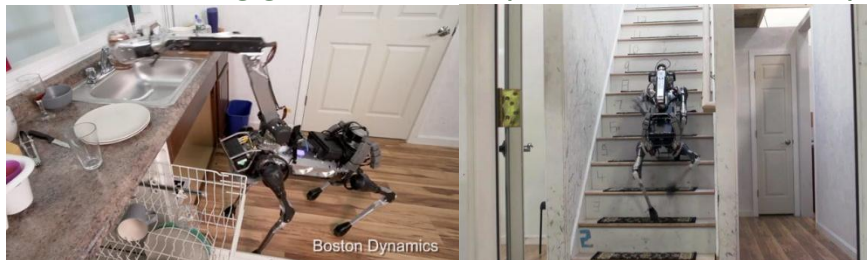


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- Legged robots (Boston Dynamics)



SpotMini

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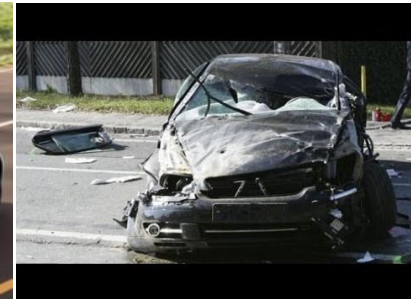
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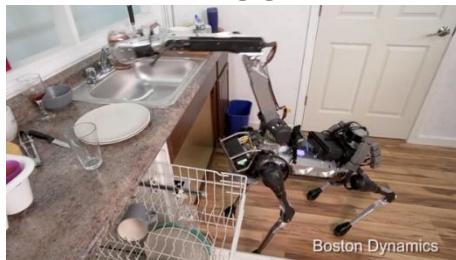


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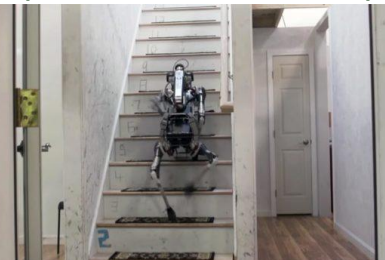


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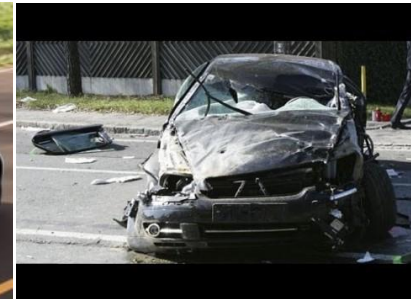
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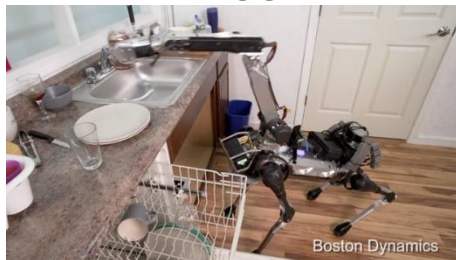


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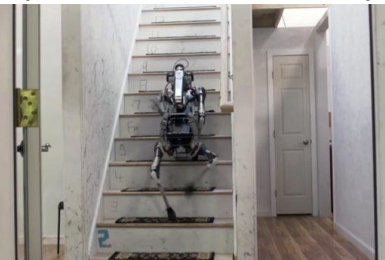


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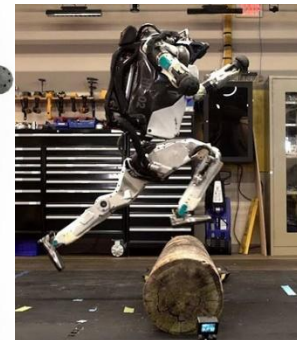
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Handle

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- Autonomous planning
 - ◆ NASA's Remote Agent program controlled the scheduling of operations for a space craft (2000).
 - ◆ Uber and Google Maps plan optimal routes for hundreds of millions of users.
- Speech recognition
 - ◆ Flight booking via conversation with an automated system
 - ◆ Real-time speech-to-speech translation
- Image understanding
 - ◆ ImageNet object recognition
 - ◆ Image captioning
 - ◆ Face recognition (boarding airplanes, bullet trains, etc.)

} Extensive use of convolutional neural networks (CNNs)
- Natural language processing
 - ◆ ChatGPT by OpenAI allows user to steer and refine a conversation (2022)

Precautions, Downsides & Failures

- Has artificial intelligence become alchemy?

Science, 4 May 2018, Vol 360, Issue 6388, p. 478

- ♠ Lack of rigor
- ♠ No explanation why some algorithms work and why others don't.
- ♠ Inability to replicate results by others

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- ♠ AI's turbulent past and uncertain future (its boom-and-bust cycle)
- ♠ Is Deep Learning deep enough?
- ♠ AI failures: briittleness, embedded bias, catastrophic forgetting, explainability, common sense etc.

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AI is not a science. But it is dramatically changing our lives.